Swami Keshvanand Institute of Technology, Management &Gramothan, Jaipur

Teaching and Examination Scheme
I Year I Semester: B.Tech. (CSE)
(Common to all UG Programmes)
Syllabus

Name of the Programme: B.Tech. in Computer Science & Engineering	**Year**: I	**Semester**: I
Course Name: Computational thinking and Programming	**Course Code**:	**Credit**: 2
Max Marks: 100	**CIE:** 40	**SEE:** 60
End Term Exam Time: 3 Hrs	**Teaching Scheme:** 2L+0T+0P	

Module No.	Contents	Hours
1	**Introduction:** Objective, Scope, Outcome of the Course and Prerequisite	1
2	**Introduction to Computers:** Computer and types of computers, Generations of computer, architecture of computer, Storage device- Primary and secondary memory, Access methods of memory, Application software v/s System software. **Introduction to Programming Language**: History and Features of Programming Language, Programming Paradigms, Types of translators, steps for compilation and execution process, Syntax and Semantic errors, Algorithms, Flowcharts/ Pseudo codes.	6
3	**Introduction to C programming language:** Introduction to C, Tokens, Data types, Operators. Flow of Control Conditional statements- If-else, Switch-case constructs, Loops- While, do-while, for.	5
4	**Functions and Pointer:** Introduction to Function, Types of functions- Predefined and user defined functions, Parameter passing functions, Recursion. Pointer Declaration and access of Pointer Variables, void Pointer, Pointer to Pointer, Pointer Arithmetic, call by value vs call by reference.	5
5	**Arrays and string:** One dimensional, Two-Dimensional Arrays and their operations, strings: Declaration, initialization, Accessing, passing arrays and strings as parameters to functions, accessing elements of arrays and string using pointer variable.	5
5	**Structure and union:** Declaration, Initialization of structure and union, passing structure to function, Use of pointers with structures. Nested structure, array of structure. **File Handling:** files and its types, file modes, operations on files (open, read, write, append, and close).	6
	Total	**28**

Text Books:

1. Programming In Ansi C|8th Edition ,2019.

Reference Books:

1. **Let Us C, Yashwant Kanetkar, Infinity Science Press,19th edition, 2022**
2. **Schaum's Outline of Programming with C, Byron S Gottfried, TATA McGraw Hill, edition, 1996.**
3. **The C Programming Language, Brian Kernighan & Dennis Ritchie, Prentice Hal,2nd edition 1988.**